

Preliminary Feasibility of Using Crowdworkers to Identify Visual Triggers for Photosensitive Epilepsy in Digital Media

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Abstract

Social media is a prominent aspect of people's everyday life but remains a potentially dangerous experience for people with photosensitive epilepsy, a condition in which exposure to certain visual triggers may induce seizures and other adverse effects. To explore alternatives for existing rule-based algorithms for detecting visual triggers in media, we conducted a preliminary interview study with 4 participants with photosensitive epilepsy and an initial experiment with 49 participants to determine the feasibility of crowdsourcing in the detection of content that may trigger seizures. We compared the crowdsourced responses with the labels of PhotosensitivityPal (a tool with high accuracy in detecting and classifying visual triggers), which indicated that crowdsourcing seizure trigger risk identification was overall feasible. We provide design opportunities for human-powered accessibility, both in the implementation of tools and in the improvement of crowdsourced detection.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Accessibility technologies**; *Accessibility systems and tools*.

Keywords

Photosensitive epilepsy, crowdsourcing, accessibility

ACM Reference Format:

Angela Davis, Connie W. Chau, and Maia Jacobs. 2025. Preliminary Feasibility of Using Crowdworkers to Identify Visual Triggers for Photosensitive Epilepsy in Digital Media. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720058>

1 Introduction

Over the past decade, the landscape of social media has changed dramatically. Following the rise of TikTok's prevalence among youth [13], short video content has grown in overwhelming popularity across all platforms. Major content-creation platforms have created their own versions of video snippets. Instagram "Reels" have

seen explosive popularity since its inception in 2020, and YouTube "Shorts" have reported a user base of over two billion [14]. Despite the prevalence of social media today, there exist various communities in which navigating online spaces poses a significant risk. Individuals with photosensitive epilepsy are one such community and are particularly vulnerable to health hazards posed by the emphasis on short video content.

Photosensitive epilepsy (PE) is a type of epilepsy in which seizures or other side effects (headaches, dizziness, and nausea) can be triggered visually. The most common form of these triggers come in (1) flashes, (2) transitions from/to the color red, and (3) patterned images. These visual triggers can be found in GIFs and videos on social media. South et al. predict that 6.65% to 9.85% of GIFs online contain potential seizure risks [17], but accessibility measures have not caught up with making online content safe for individuals with PSE. In other words, social media usage and visual trigger risk have risen over the past decade, yet insufficient work has been done to make social media a safe experience for individuals with PSE. Notably, a substantial portion of today's youth (aged 13-17) are social media users [13], and individuals from the age 11 to 20 years old are also the most likely to show signs of photosensitivity [9]. Thus, developing ways to make social media a safer space for people with PSE is a necessary step for increasing the accessibility and safety of online experiences for youth and young adults.

We first conducted semi-structured interviews to gain a preliminary understanding of how individuals with PSE navigate online spaces and the type of support they desired for avoiding online visual triggers. Importantly, we found that current systems did not meet users' needs, which motivated us to explore solutions beyond existing algorithmic detection of visual triggers. Previous research on utilizing crowdsourcing for improving accessibility have proven to be successful [2], indicating that it may also be an option in identifying seizure-inducing risks in visual content on social media. Therefore, we conducted a preliminary experiment study with 49 participants to gauge the feasibility of crowdsourcing in visual trigger detection by comparing crowdsourced workers' abilities to accurately identify triggers in GIFs with that of PhotosensitivityPal [17]. *Our initial experiment begins to explore the research question: "Can humans with minimal training detect the most common visual triggers of photosensitive epilepsy?"*

Our work contributes towards a **nascent research agenda for using crowdsourcing for PSE visual triggers, including preliminary data on its performance in potentially seizure-inducing media and perspectives of crowdworkers** towards integrating this identification method more sustainably.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720058>

2 Related Work

2.1 Visual Trigger Types

According to the Epilepsy Foundation, a flash on the screen of an electronic device must meet three qualifying criteria to be considered a risk: a change in luminance greater than or equal to 20 cd/m²; a frequency greater than or equal to 3 Hz; and an occupancy of a solid visual angle greater than or equal to 0.006 steradians (which is roughly equivalent to 25% of a screen) [9]. Common examples of flashes in visual effects are flickering lights or strobe lights; however, even seemingly benign components of video content, such as fast camera movement, can meet the aforementioned requirements. A red transition/flash, regardless of luminance, is considered a seizure risk if it contains a transition from/to a saturated red (i.e., red transition). Repeated pattern triggers consist of five light–dark pairs of stripes, last on screen for longer than 0.5 seconds, occupy 25% or more of the screen, and have light stripes greater than 50 cd/m² in luminance are considered a visual trigger for individuals with PSE [19]. In this paper, we explore how well crowdworkers with minimal training can identify each of these three visual triggers.

2.2 Existing Work for Photosensitive Accessibility

EpilepsyBlocker and PhotosensitivityPal are both browser extensions that analyze GIFs and other forms of media as the user browses a website, and provide an evaluation on whether the examined content poses a potential seizure risk. Creator-driven and consumer-driven efforts have been focused on the implementations and improvements of algorithmic/computational identification (i.e. "rule-based") systems. Our work, on the other hand, aims to offer a new perspective on the feasibility of human detection. Performance metrics of some existing solutions have been studied in previous work by South et al. [17], using several datasets consisting of GIFs containing no triggers, flashes, red transitions, and patterns. Three solutions were tested: PEAT, PhotosensitivityPal, and EpilepsyBlocker (metrics for the last tool were not considered as it was not able to recognize any GIFs in any of the datasets). The results of the study showed high accuracy rates for PhotosensitivityPal (94% accuracy in detecting flashes, 100% accuracy in detecting red transitions, and 97% accuracy in detecting repeated patterns), as well as a high percentage of label agreement between PhotosensitivityPal and PEAT in their classifications of GIFs (e.g., for a given GIF that was analyzed by both PhotosensitivityPal and PEAT, both tools correctly agreed that it contained a flash trigger). We extend South et al. [17]'s work by utilizing a subset of this dataset to determine the results of how much the humans' classification of GIFs agree with PhotosensitivityPal's and provide a different approach to traditional, rule-based systems via crowdsourcing instead.

2.3 Crowdsourcing

Crowdsourcing is a methodology that involves utilizing human power for completing specified tasks [7], and it has been frequently explored in domains such as metric acquisition, design improvements, and digital accessibility [6, 15, 16]. For online accessibility specifically, Bigham et al. argues that human computation serves as a powerful tool in serving communities with differing needs [2].

Examples of previous research include quantifying video quality [8], locating bus stops and landmarks [11], and creating thumbnail captions [1]. In terms of an alternative for computational media labeling, Von Ahn and Dabbish [18] provides a novel system of using gamified crowdsourcing to label images on the Web. He suggests in his study that complex computer vision problems can be solved by human power and subsequently, accessibility can be improved. In this paper, not only do we explore crowdsourcing in a different domain (i.e., digital accessibility for individuals with PSE), but also offer the perspective of crowdworkers and their thoughts, opinions, and perceived contributions to the task of identifying visual triggers intended to support other people.

3 Study 1: Interviewing Individuals with Photosensitive Epilepsy

3.1 Methods

We conducted a semi-structured user interview study and recruited 4 participants through ResearchMatch. The criteria for participant eligibility was being 18 years of age or older, proficient in English, residing in the U.S., having been diagnosed by a medical professional with PSE, and owning or having previously owned a social media account. Interviews were conducted remotely via Zoom. The interview study received exempt status from our institution's IRB due to minimal to no risk of visual trigger to participants as this protocol focused primarily on discussing people's experiences with PSE and social media. During the interview, participants were asked questions centered around three central themes: their history with PSE, their social media usage, and their interests/wants for support tools. For more details, see Appendix A. All participants were white and female, ranging from 26 to 40 years of age (see Table 1).

3.2 Study 1 Findings

3.2.1 Current Strategies for Avoiding Visual Triggers in Online Spaces. Participants shared that they used both measures within social media platforms and physical or external methods to avoid visual triggers online. Within platforms, the most frequent response was turning off the autoplay feature, if possible. Although this was an option that would only give users autonomy in choosing which videos they wanted to watch, it was described as a "*huge game changer*" by P1. Managing the content the participants followed and received was also a popular strategy. Unsurprisingly, they avoided accounts that posted flashy content inundated with visual effects. However, participants themselves acknowledged that this was not guaranteed to protect them from visual triggers and even expressed how even videos sent by their friends were also not guaranteed as risk-free. As for external measures, a common method shared by all participants was to simply look away when a visual trigger appeared, though this was sometimes not sufficient. All four participants were unaware that external tools for identifying visual triggers in online content (i.e., PhotosensitivityPal or PEAT) existed.

3.2.2 Lack of Public Awareness of Photosensitive Epilepsy. Participants frequently voiced their frustrations with how inaccessible social media was for individuals with PSE. P4 mentioned that a part of the issue may be **due to the lack of awareness and empathy for the obstacles people with PSE face**. This sentiment is

Table 1: Demographic Data of Participants

Participant ID	Age	Social Media Platforms
P1	26	Facebook, Instagram, YouTube, TikTok, Snapchat
P2	27	Instagram, Facebook, TikTok, YouTube
P3	38	YouTube, Facebook
P4	28	Twitter, Instagram, Facebook, YouTube, TikTok

further consolidated by P1’s statement earlier regarding how even their own friends, who have known about their condition for years, sometimes forget about the danger visual triggers pose to them.

3.2.3 Desired Design Features for Accessibility Tools. Participants were asked to describe features for an ideal tool for detecting visual triggers in social media content. Firstly, all participants expressed that the **non-obstructive integration of such a tool would be crucial in their online experience**, with P2 saying, “*I’d want for [the tool] to keep up with me*” as they browsed websites (rather than the tool preventing the website from loading prior to viewing), and alluded to the tool being a platform-driven solution “*from the major social media companies [...that] incorporate features that either you can turn on if you want to, if you have epilepsy*” (P4). Secondly, participants desired transparency through **listing of all the visual triggers in the content** that could be considered a seizure risk. Participants acknowledged that different visual triggers yielded different responses from individuals with PSE, **expressing a need for tools to also categorize and communicate to users what types of visual triggers may be present**. P1 also offered a new perspective in that there were times in which they were more susceptible to seizures or other health conditions, so a visual trigger that would not normally be of significant concern could become something they wanted to actively avoid. Being told why a video may be dangerous for them would allow users the flexibility to make their own decisions, and a more descriptive warning would be better. Finally, participants also wanted the **option to view potentially triggering media**, with two citing how it would be helpful for detailed information like when a visual trigger would appear in especially longer-form content.

4 Study 2: Crowdsourcing Feasibility Experiment

Our findings in the interview study motivated the crowdsourcing feasibility experiment. We found that participants’ needs were not being met by current social media platforms nor did they know that tools like PhotosensitivityPal were even available to them. A platform-driven solution was suggested by a participant and seemed to be the best fit to respond to their strong desire for a seamless social media experience, which was not yet possible with current accessibility tools. We considered existing pipelines for platform-based, community-driven content moderation, such as reporting posts on Instagram [3] and community notes on X (formerly known as Twitter) [5], and the potential limitations in scalability for traditional algorithmic detection of visual triggers. Consequently, we were motivated to conduct an initial experiment to see if crowdsourcing could be a viable alternative.

4.1 Methods

4.1.1 Participants and Eligibility. 49 participants were recruited through Prolific¹. Eligibility criteria was being 18 years of age or older, proficient in English, residing in the U.S., and most importantly, *not having* PSE, any health conditions triggered by visual stimuli, and any history of having a visually-induced seizure. The experiment was granted exempt status by our institution’s IRB because our screening criteria explicitly excluded individuals who had (or potentially had) PSE and our protocol included measures to minimize health risk (see Limitations for more details). Participants were compensated \$15 for completing the experiment via a Qualtrics survey.

4.1.2 Dataset. The GIFs utilized for our experiment were chosen from the social media study dataset provided by South et al. [17] in their supplementary materials. This dataset consisted of 2000 GIFs collected from social media platforms (i.e., Twitter, Tumblr, Tenor, and GIPHY), and PhotosensitivityPal’s results or “labels” for each GIF in the dataset (i.e., if trigger was present and if so, which of 3 trigger types). We use this dataset because it contains GIFs from social media websites (including those without visual triggers), reflecting actual content that a user may encounter while browsing online. Although the labels for this dataset were provided by PhotosensitivityPal and were subsequently not ground truth labels, it was not feasible to get ground truth labels in totality and PhotosensitivityPal’s labels were the next appropriate alternative due to its high performance and accuracy. This dataset was used for both the entirety of the experiment (training and experiment module).

4.1.3 Training & Experiment Modules. Participants completed a mandatory training module that included definitions and examples of each visual trigger, as well as a 3-question pre-assessment that required all correct responses before moving onto the full experiment module. For the experiment, 39 GIFs were randomly selected from the dataset (10 GIFs for each visual trigger, except for red transitions, in which only nine GIFs were available) and 10 GIFs without any seizure risks. 5 GIFs would then be randomly chosen for each visual trigger to be used in each iteration of the experiment. We randomly selected GIFs because an initial pilot test of the experiment revealed that certain GIFs were much more difficult to analyze than others, resulting in skewed results for identification. We also manually reviewed the 39 GIFs in our subset of South et al.’s dataset to ensure there were no deleted or inactive links. For each GIF, participants were first asked if they thought a visual trigger was at all present. If participants indicated “Yes”, they would answer a follow-up question on which of the three visual triggers

¹<https://www.prolific.com/>

was present (see Figure 3). The rationale behind testing the ability to identify specific triggers was inspired by the findings in the interviews from Study 1, where the interviewees emphasized the significance of being informed on which specific types of visual triggers may be present in a social media post. Every five questions were followed by an optional free response question asking the participant about their thoughts on the experiment thus far. This was intended to give participants a break from analyzing visual stimuli and to also elicit their perspectives as crowdworkers for these tasks.

4.2 Experimental Results

We calculated the percentage of the human responses that matched the labels provided by PhotosensitivityPal by dividing the number of matched responses by the total number of responses—we refer to this as a "similarity score". Recall that because ground truth was not feasible, our results are in comparison to PhotosensitivityPal. When we state that participants "correctly" agreed with PhotosensitivityPal, we mean they had the same labels/responses (i.e., "yes/no" to whether a given GIF has a visual trigger; and if "yes", further identifying if the trigger is a red transition, flash, or pattern). **Overall similarity:** Of the 931 responses, 681 correlated with the PhotosensitivityPal's identification as either safe or containing a visual trigger. This means that 73.1% of participant responses agreed with PhotosensitivityPal's labels of whether a GIF was a potential seizure risk or not (see Figure 1). **Similarity score by trigger type:** Individual label matching was evaluated by trigger type. For red transitions, participants agreed with PhotosensitivityPal on the classification of 51.5% of the GIFs; for patterns, it was 45.3%; and for flashes, it was 84.4% (see Figure 1).

4.2.1 Themes of Crowdsourcing Perspectives & Feedback. The majority of participants expressed that **identification was difficult and they were often uncertain** about what they were doing. Some simply said that the visual triggers were "*hard to distinguish*" and that they were "*questioning how well*" they were performing, which strongly seemed to be due to a lack of knowledge regarding how present visual triggers needed to be in a GIF to classify as a seizure risk. This led to the theme of **needing both more thorough training and more training in general**. Some comments that mentioned the difficulty of identification also added that they would have liked more practice. Finally, involvement in this study **brought more awareness to the potential seizure risks** that were present in online content. Many participants expressed astonishment as to how frequently visual triggers appeared. P37 stated that "*Once educated on the various triggers, I was kind of surprised how ubiquitous they are in media*". This realization seemed to also inspire discussion and empathy for the PSE community, as E52 mentioned thinking about how "*a lot of these are GIFs which people commonly send each other via messages or post to social media*" and that they "*never thought about visual triggers for people who experience seizures*". E40 went a step further in their discovery of how many things in media have patterns and red transitions, saying that "*it must feel debilitating if you have a sensitivity to either*". Our inclusion of responses from crowdsourced workers in our experiment demonstrated how they fostered empathy and a desire to learn more about the experiences of those who have PSE.

5 Discussion

In this crowdsourcing feasibility study, we explore the correlation between the labels provided by existing visual trigger detection solutions to humans with minimal training. Our findings have implications for areas of success in human identification where crowdsourcing can be best leveraged to improve social media accessibility.

Our research question asked if humans with minimal training could identify visual triggers at all; we found that they could. When detecting whether or not a GIF contained a visual trigger and thus considered a potential seizure risk, our participants agreed with PhotosensitivityPal on 73.1% of the GIFs (i.e., meaning that participants correctly agreed with PhotosensitivityPal that the GIF had a visual trigger). These results are significant because of PhotosensitivityPal's high accuracy rates, which was the closest alternative to ground truth for this preliminary study. For each visual trigger type, participants provided answers that were the closest to PhotosensitivityPal's classification for flashes (84.4%), followed by a large gap for red transitions (51.5%) and then patterns (45.3%). These results suggest that humans can identify flashes with reasonable precision, but are worse at identifying red transitions and patterns. It is worth noting that flashes are the predominant/most common seizure risk for individuals with PSE [9], and as such, methodologies to use crowdsourcing for flash detection may have a meaningful impact. Additionally, according to experiment feedback, participants believed that a potential challenge in identifying the other triggers was partly due to a lack of knowledge of how intense a visual trigger needed to be in order to classify as a potential seizure risk. This implies opportunities for improvement for the training module that may be able to improve people's ability to correctly identify these other two trigger types. However, in conjunction with these findings and the many potential pitfalls in using only crowdsourcing for moderating social media posts [10], implementations of crowdsourcing specifically for PSE should also consider the integration of automated systems.

Our findings from the user interviews that motivated the crowdsourcing feasibility study also indicated that users place a heavy emphasis on an accessible system that was unobstructive to their social media experience (Section 3.2.1). To address this user need, we envision a platform-driven solution that utilizes a combination of user-generated reporting and an automated, collaborative human-AI review system to reduce seizure risk exposure to individuals with PSE. Machine detection can complement human identification and support people with tasks that humans find more complicated (i.e., detecting red transitions and patterns). Similar implementations of content moderation on popular social media platforms have seen success. Instagram, for example, utilizes a combination of artificial intelligence (AI) and human reviewers to analyze or take down posts that violate their community guidelines [3]. X (formerly known as Twitter) uses a combination of machine learning and a crowdsourcing network of users for their "community notes" feature, which aims to combat misinformation and add additional context to the posts on their application [5]. Taking into consideration the success of these content moderation systems and the results of our study, we see an opportunity for social media platforms to leverage these existing pipelines to improve the accessibility of

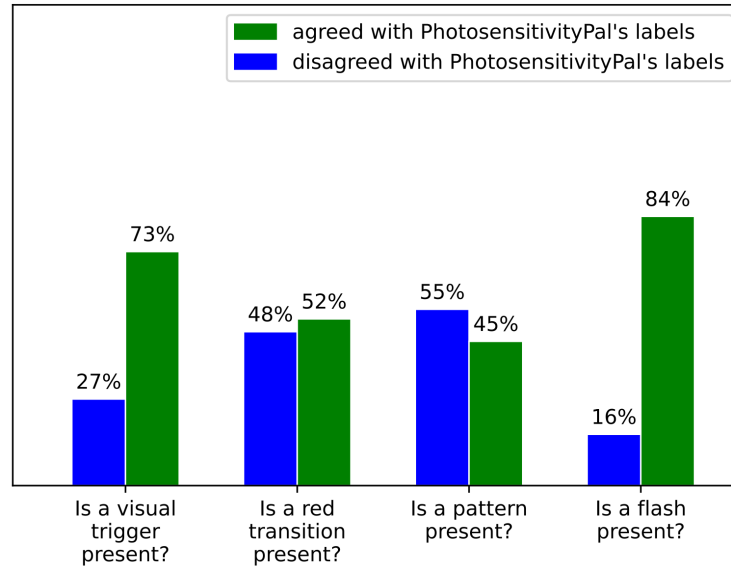


Figure 1: Percentages of participants that agreed and disagreed with PhotosensitivityPal's labels for if a GIF contained a visual trigger (total number of responses) and if a GIF contained a specific trigger (percentage of responses for each trigger type)

their platforms to those who are sensitive to visual triggers. While our study offers preliminary data on the performance of humans with minimal training detecting visual triggers, future work could explore the validation of crowdsourcing with a simulated interface of a common social media app and on a mobile device (where most people would engage in social media content).

Another implication we found was that crowdsourcing as a methodology can have the additional benefit of raising awareness and empathy for the benefactors of their tasks. Our findings showed that as participants worked through the experiment, they related what they learned about visual triggers to their own experiences on social media. They developed empathy for users who were sensitive to the visual stimuli they were identifying under the consideration that these triggers seemed omnipresent in current Internet culture (Section 4.2.3). This is particularly meaningful as interview participants felt that a current lack of empathy and knowledge on photosensitive epilepsy may play a part in the lack of accessible tools and online experiences for individuals sensitive to visual stimuli (Section 3.1.1). Existing literature affirms that crowdsourcing can become successful as a means to increase awareness on a certain topic (e.g., Internet privacy) [4], and that crowdsourcing can bring awareness of stigmatized issues, topics, or experiences to crowdworkers that they may have never engaged with otherwise [12]. Social media is a place where many users receive news, but it can also be a space to reinforce empathy. The integration of crowdsourcing could reduce barriers to understanding the lived experiences of the PSE community and the risks they have been forced to navigate online. Future work can take a human-centered approach to further

explore how their work as crowdsourcing workers in visual trigger detection has impacted their experience on social media.

6 Limitations

For both the user interviews (Study 1) and the crowdsourcing feasibility experiment (Study 2) study, there were sampling limitations. For Study 1, all participants for the user interviews were white, female, possessed at least a high school diploma, were medically diagnosed with PSE, and lived in the United States. We acknowledge that our small sample size of 4 is not representative of all individuals with PSE, especially as individuals who are unable to be officially diagnosed with PSE may be at greater risk when navigating online spaces. Also, users in other countries may interact with social media differently or use different social media platforms that were not mentioned in this study. Due to the lack of responses from more diverse communities during recruitment, the voices of underserved populations in particular were not included in our analysis of how crowdsourcing may meet potential design requirements in ways that existing solutions do not. A critical area for future work would be comparing and contrasting these experiences, being conscientious of how each participants' identity as someone with PSE intersects with other aspects of their lived experience.

For Study 2, participants identified as predominantly white and cisgender, and all resided in the United States. We acknowledge that this is not representative of all social media users without PSE and emphasize that with a sample size of 49 participants, this only offers a preliminary understanding of how the average person with minimal training might perform in identifying visual triggers in

comparison to existing solutions. We encourage future research to conduct our experiment on a much greater scale with a larger population of participants to provide more generalizable data on using crowdsourcing to identify visual triggers. Furthermore, in the absence of ground truth labels, we compare crowdsourced visual trigger identification with PhotosensitivityPal's labels for our experiment study. We acknowledge that, despite PhotosensitivityPal's high accuracy, all algorithmic systems have their own biases, and urge future work to rigorously determine appropriate ground truth measures for future comparative work. Additionally, we reiterate that our results only provide preliminary data on the performance of crowdworkers relative to existing algorithmic identification tools. With regards to our dataset, GIFs that had multiple visual triggers were classified as having only one visual trigger. However, only two GIFs in our dataset of 38 GIFs had multiple visual triggers so we do not believe this had a significant impact on our results.

Additionally for Study 2, we communicated clearly that the criteria for participation was to (1) not have PSE, (2) any other health condition that would make one susceptible to visual triggers, or (3) any history of experiencing a visually-induced seizure. This statement was clearly written in our experiment description and the first page of the consent form in the Qualtrics survey. As an added precautionary measure, participants completed a demographic survey portion that included a question to check if they had PSE (or experienced symptoms akin to PSE) prior to any actual study material, including the training and experiment modules). If a participant answered "Yes" to this question, the survey immediately ended and they were not allowed to proceed with the study. Future work following our direction of using crowdsourcing for PSE trigger identification should take further conscientious precautions to maximize the safeguards for crowdworkers who may potentially be impacted by visually-triggering content. One suggestion would be to conduct the experiment in-person with medical staff readily available so that participants may be supervised and receive immediate care in case of an adverse reaction. We encourage more thorough assessment of potential risks to participants, including evaluating a potential participant's experiences with PSE, visual triggers, and/or seizures prior to any study activities. This may come in the form of describing alternative ways to account for potential unfamiliarity with PSE (e.g., asking if participants have experienced symptoms such as dizziness when looking at particular media) or even diagnostic exams.

7 Conclusion

In this paper, we presented an initial feasibility study on how humans with minimal training perform at identifying potential seizure-inducing visual triggers on social media GIFs, in comparison to existing rule-based machine solutions. Through user interviews, we found that there were still many barriers in social media that made the user experience inaccessible for individuals with PSE and reported design requirements for future implementations of visual trigger detection systems. By conducting a crowdsourcing feasibility study for human identification, we discovered that humans often agreed with PhotosensitivityPal's labels in the detection of potentially seizure-inducing GIFs. For the three visual trigger types we tested (flashes, red transitions, and patterns), we found

that humans were best at identifying flashes—the most common seizure risk for individuals with photosensitive epilepsy. Participants also expressed that they could perform better given additional training, which suggests that improvements in the training provided may result in greater success in identifying the other two trigger types they did not detect as well. Based on our findings from both the interview and crowdsourcing experiment studies, we shared implications for the design of platform-driven systems utilizing human-AI collaboration, along with feedback from participants that indicate how crowdsourcing work could be an avenue to build empathy for crowdworkers on the experiences of social media users with photosensitive epilepsy. This work contributes a preliminary understanding of how crowdsourcing as a methodology may be feasible in improving digital accessibility for individuals with photosensitive epilepsy. We motivate future research on alternate approaches to seizure risk detection in social media content and increasing awareness of the risks the photosensitive epilepsy community face online.

References

- [1] Carlos Aguirre, Shiye Cao, Amama Mahmood, and Chien-Ming Huang. 2023. Crowdsourcing Thumbnail Captions: Data Collection and Validation. *ACM Transactions on Interactive Intelligent Systems* 13, 3 (Sept. 2023), 1–28. doi:10.1145/3589346
- [2] Jeffrey P. Bigham, Richard E. Ladner, and Yevgen Borodin. 2011. The design of human-powered access technology. In *The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility*. ACM, Dundee Scotland, UK, 3–10. doi:10.1145/2049536.2049540
- [3] Instagram Help Center. [n.d.]. How Instagram uses artificial intelligence to moderate content | Instagram Help Center. <https://help.instagram.com/423837189385631>
- [4] Ioannis Chrysakis, Giorgos Flouris, Maria Makridaki, Theodore Patkos, Yannis Roussakis, Georgios Samaritakis, Nikoleta Tsampanaki, Elias Tzortzakakis, Elisjanna Ymeralli, Tom Seymoens, Anastasia Dimou, and Ruben Verborgh. 2021. A Rewarding Framework for Crowdsourcing to Increase Privacy Awareness. In *Data and Applications Security and Privacy XXXV (Lecture Notes in Computer Science)*, Ken Barker and Kambiz Ghazinour (Eds.). Springer International Publishing, Cham, 259–277. doi:10.1007/978-3-030-81242-3_15
- [5] X Corp. [n.d.]. Community Notes. <https://communitynotes.twitter.com/guide/en/about/introduction>
- [6] John J. Dudley, Jason T. Jacques, and Per Ola Kristensson. 2021. Crowdsourcing Design Guidance for Contextual Adaptation of Text Content in Augmented Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–14. doi:10.1145/3411764.3445493
- [7] Enrique Estellés-Arolas, Raúl Navarro-Giner, and Fernando González-Ladrón-de Guevara. 2015. Crowdsourcing Fundamentals: Definition and Typology. In *Advances in Crowdsourcing*, Fernando J. Garrigos-Simon, Ignacio Gil-Pechuán, and Sofia Estelles-Miguel (Eds.). Springer International Publishing, Cham, 33–48. doi:10.1007/978-3-319-18341-1_3
- [8] Oscar Figuerola Salas, Velibor Adzic, Akash Shah, and Hari Kalva. 2013. Assessing internet video quality using crowdsourcing. In *Proceedings of the 2nd ACM international workshop on Crowdsourcing for multimedia*. ACM, Barcelona Spain, 23–28. doi:10.1145/2506364.2506366
- [9] Robert S. Fisher, Jayant N. Acharya, Fiona Mitchell Baumer, Jacqueline A. French, Pasquale Parisi, Jessica H. Solodar, Jerzy P. Szaflarski, Liu Lin Thio, Benjamin Tolchin, Arnold J. Wilkins, and Dorothée Kasteleijn-Nolst Trenité. 2022. Visually sensitive seizures: An updated review by the Epilepsy Foundation. *Epilepsia* 63, 4 (2022), 739–768. doi:10.1111/epi.17175
- [10] Arpita Ghosh, Satyen Kale, and Preston McAfee. 2011. Who moderates the moderators?: crowdsourcing abuse detection in user-generated content. In *Proceedings of the 12th ACM conference on Electronic commerce*. ACM, San Jose California USA, 167–176. doi:10.1145/1993574.1993599
- [11] Kotaro Hara, Shiri Azenkot, Megan Campbell, Cynthia L. Bennett, Vicki Le, Sean Pannella, Robert Moore, Kelly Minckler, Rochelle H. Ng, and Jon E. Froehlich. 2013. Improving public transit accessibility for blind riders by crowdsourcing bus stop landmark locations with Google street view. In *Proceedings of the 15th international ACM SIGACCESS Conference on Computers and Accessibility*. ACM, Bellevue Washington, 1–8. doi:10.1145/2513383.2513448
- [12] Mikaela Hildebrand, Claudia Ahumada, and Sharon Watson. 2013. Crowd-OutAIDS: crowdsourcing youth perspectives for action. *Reproductive Health*

- Matters* 21, 41 (May 2013), 57–68. doi:10.1016/S0968-8080(13)41687-7
- [13] Emily A. Vogels Massarat, Risa Gelles-Watnick and Navid. 2022. Teens, Social Media and Technology 2022. <https://www.pewresearch.org/internet/2022/08/10/teens-social-media-and-technology-2022/>
 - [14] Romana Pawar. 2023. YouTube is redesigning the streaming experience for viewers and advertisers. <https://blog.google/products/ads-commerce/youtube-is-redesigning-the-streaming-experience-for-viewers-and-advertisers/>
 - [15] Judith Alice Redi, Tobias Hoffeld, Pavel Korshunov, Filippo Mazza, Isabel Povia, and Christian Keimel. 2013. Crowdsourcing-based multimedia subjective evaluations: a case study on image recognizability and aesthetic appeal. In *Proceedings of the 2nd ACM international workshop on Crowdsourcing for multimedia*. ACM, Barcelona Spain, 29–34. doi:10.1145/2506364.2506368
 - [16] Shuyi Song, Jiajun Bu, Ye Wang, Zhi Yu, Andreas Artmeier, Lianjun Dai, and Can Wang. 2018. Web Accessibility Evaluation in a Crowdsourcing-Based System with Expertise-Based Decision Strategy. In *Proceedings of the 15th International Web for All Conference*. ACM, Lyon France, 1–4. doi:10.1145/3192714.3192827
 - [17] Laura South, David Saffo, and Michelle A. Borkin. 2021. Detecting and Defending Against Seizure-Inducing GIFs in Social Media. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3411764.3445510
 - [18] Luis Von Ahn and Laura Dabbish. 2004. Labeling images with a computer game. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, Vienna Austria, 319–326. doi:10.1145/985692.985733
 - [19] Arnold Wilkins, John Emmett, and Graham Harding. 2005. Characterizing the Patterned Images That Precipitate Seizures and Optimizing Guidelines To Prevent Them. *Epilepsia* 46, 8 (2005), 1212–1218. doi:10.1111/j.1528-1167.2005.01405.x

A Appendix

A.1 Images from Qualtrics Experiment

A.2 Interview Questions

A.2.1 General Health Status.

- (1) How long have you been diagnosed with photosensitive epilepsy?
- (2) How often do you experience seizures?
- (3) Generally, what have you found to have been the worst triggers for you?
- (4) Has social media content ever triggered a photosensitive epileptic seizure for you? Can you talk about it?
- (5) At any point has your doctor talked to you about social media/digital content risk?
 - (a) If they have, what were the strategies in decreasing risk for digital content (including movies and television shows) that they recommended?

A.2.2 Personal Social Media.

- (1) Can you walk me through a typical day in how you use social media?
 - (a) Who do you follow?
 - (b) When do you use social media?
 - (c) Which types of social media do you use?
 - (d) How often do you use each one of those social media platforms?
 - (e) Is your feed only photos with no videos?

A.2.3 Social Media Use.

- (1) How do you currently deal with the risk involved in using social media?
 - (a) Are there any commercial software or features that you have used?
 - (b) How effective would you say these strategies are?
 - (c) Are there any other behaviors/things you do to avoid triggers on technology?

- (2) In terms of your photosensitive epilepsy, is there an increased risk in watching television/movies than there is using your phone or computer?

A.2.4 Future Technology Interest/Use.

- (1) On building software that can detect videos containing visual triggers on social media - how would you imagine that working?
 - (a) What features or controls would that have?
 - (b) Would you want a text warning? The video to be skipped?
- (2) Would you need it more on your phone or computer?
- (3) How interested would you be in the technology that could result from this study?



What visual trigger is present in the above video?

☐ Flash/Fast Transition

☐ Red Transition

☐ Pattern

What visual trigger is present in the above video?

☐ Flash/Fast Transition

☐ Red Transition

☒ Pattern

Nice try. Try paying attention to the background of the video and the colors involved. Please try again.

Figure 2: Example practice question from the training module and explanation for incorrect answer



Visual Trigger Definitions:

Flash/Fast Transition: a pair of opposing changes in brightness or a quick change in brightness

Red transition: any pair of changes of color and brightness involving a saturated red

Pattern: contains clearly discernible stripes, numbering more than five light-dark pairs of stripes in any orientation

Does this video contain a visual trigger?

☐ Yes, it contains a visual trigger

☐ No, it does NOT contain a visual trigger

Does this video contain a visual trigger?

☒ Yes, it contains a visual trigger

☐ No, it does NOT contain a visual trigger

Which visual trigger(s) is (are) present?

☐ Flash/fast transition

☐ Red transition

☐ Patterns

Figure 3: Example practice question from the experiment module, including the drop down to select a visual trigger